

```

1 def candidate_elimination(data):
2     num_attributes = len(data[0]) - 1
3
4     # Initialize Specific and General boundaries
5     S = ['0'] * num_attributes
6     G = [['?'] * num_attributes]
7
8     # Initialize S with the first positive example
9     for row in data:
10         if row[-1] == 'Yes':
11             S = row[:-1].copy()
12             break
13     for row in data:
14         inputs, label = row[:-1], row[-1]
15
16         if label == 'Yes':
17             # Handle positive example
18             for i in range(num_attributes):
19                 if inputs[i] != S[i]:
20                     S[i] = '?'
21             # Remove from G any hypothesis inconsistent with positive example
22             G = [g for g in G if all(g[i] == '?' or g[i] == inputs[i] for i in
23 range(num_attributes))]
24

```

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25     else:
26         # Handle negative example
27         G_new = []
28     for g in G:
29         if not all(g[i] == '?' or g[i] == inputs[i] for i in
30 range(num_attributes)):
31         G_new.append(g) # Already inconsistent, so keep it
32     else:
33         # Specialize g
34         for i in range(num_attributes):
35             if g[i] == '?' and inputs[i] != S[i]:
36                 g_candidate = g.copy()
37                 g_candidate[i] = S[i]
38         if g_candidate not in G_new:
39             G_new.append(g_candidate)
40         G = G_new
41     return S, G
42 # Example Usage
43 if __name__ == "__main__":
44     dataset = [
45         ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
46         ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
47         ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
48         ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']

```

```
[ Sunny , warm , high , strong , cool , change , yes ]  
]
```

```
s_boundary, g_boundary = candidate_elimination(dataset)  
print("Specific Boundary (S):", s_boundary)  
print("General Boundary (G):", g_boundary)
```

Program 6: Candidate Elimination

Specific Boundary (S): ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']

General Boundary (G): [['Sunny', '?', '?', '?', '?', '?'],
['?', 'Warm', '?', '?', '?', '?']]

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